

Ahmed Loughzali

✉ ahmed.loughzali@gmail.com

[in](#) LinkedIn

[GitHub](#)

[Personal website](#)

Availability: Seeking a 6-month research internship starting January/February 2027; open to international relocation.

Research interests: Embodied AI, robotics, multimodal perception (vision, audio, text) and AI for Good.

Education

Fall 2026 **Incoming Exchange Student.**

Gwangju Institute of Science and Technology (GIST), South Korea

Selected Coursework: 3D vision, Robot control, Haptics, AI for healthcare

2022 – 2027 **MSc in Computer Science.**

IMT Nord Europe (Mines-Telecom), France

Majoring in AI & Data Science.

Experience

May 2026 – Aug. **AI Researcher.**

2026 *Algoverse, Remote*

- Competitive research program (scholarship + compute grant).
- Researching the adversarial robustness of discrete image tokenizers in medical context, paper in writing.

Apr. 2026 – Sept. **AI & Robotics Research Intern.**

2026 *Enchanted Tools, Paris, France*

- Worked on the conversational audio stack of Mirokaï, a cute social humanoid robot .
- Integrated on the robot and benchmarked state-of-the-art turn-taking models (VAP, TurnGPT, SmartTurn, Semantic-VAD via Silero & custom Whisper) and explored fine-tuning Qwen-based models for multilingual end-of-turn detection.
- Built an end-to-end pipeline for barge-in detection: synthetic data generation on NVIDIA Thor, model training, and real-time deployment on Jetson Orin NX.
- Optimized models for real-time human-robot interaction under strict latency constraints (quantization), and streamlining the edge deployment lifecycle: cross-compilation, gRPC service integration, and CI/CD automation via GitLab pipelines.
- Selected to pitch the robot to enterprise clients at VivaTech and Santexpo in collaboration with Microsoft staff.

May 2025 – Aug. **AI Engineer Intern.**

2025 *Ferromobile, Paris, France*

- Built and optimized a real-time Text-to-Speech service (FastAPI, Piper TTS) on NVIDIA Jetson Xavier, achieving $\sim 0.6s$ latency with caching, integrated via TCP/HTTP to deliver voice guidance to passengers of the autonomous vehicle.
- Designed and deployed a local RAG pipeline (Docling, LangChain, ChromaDB, Mistral 7B via Ollama) with a Streamlit web interface for intelligent querying of technical documents.
- Containerized the full architecture using Docker Compose and Dev Containers to ensure reproducible and reliable deployment on embedded industrial hardware.

Projects & Research

- | | | |
|-----------------------|--|--------|
| June 2026 | Surgical World Model (EB-JEPA) | [Code] |
| | <i>World Models Hackathon, endorsed by Yann LeCun</i> | |
| | Built an action-conditioned JEPA (EB-JEPA) that predicts future latent states from wrist-camera video and proprioception, applied to surgical robotics. SLURM-managed multi-GPU distributed training on an HPC cluster (8× NVIDIA H100). | |
| Feb. 2026 | OncovIA | [Code] |
| | <i>Mistral AI x GE Healthcare Hackathon — 1st Place</i> | |
| | Multimodal oncology pipeline combining Computer Vision and Agentic RAG to help radiologists generate accurate, patient-specific follow-up reports. | |
| Oct. 2025 – Feb. 2026 | Drone Detection using Spiking Neural Networks. | |
| | <i>Research Project, IMT Nord Europe</i> | |
| | Authored a review paper on drone detection focusing on neuromorphic computing and energy-efficient SNNs. Fine-tuned spiking neural network architectures (Spiking YOLOv8, Spiking SSD) on the FRED dataset. Achieved 92% accuracy with a 69M parameter model optimized for low-power edge inference. | |
| Sep. 2024 – Apr. 2025 | Ozone Measurement System on Drone. | |
| | <i>Embedded Hardware & IoT Project, IMT Nord Europe</i> | |
| | Designed and built a low-cost (< 150€) autonomous embedded system mounted on a drone for real-time ozone measurement (ESP8266, SEN0321, GPS). Handled full hardware design, wiring, C++ embedded software, and remote server networking (Flask + InfluxDB + Grafana), achieving an 80% correlation with a 3000€ professional reference sensor. | |

Technical Strengths

Languages	Python, C++, SQL, Bash, LaTeX
Frameworks	PyTorch, Hugging Face, LangChain, FastAPI, gRPC, OpenCV, NumPy, Pandas, Scikit-learn
Tools & Infra	Linux, Git, Docker, GitLab CI/CD
Hardware & Edge	NVIDIA Jetson, ESP, Cross-Compilation, Low-Power Inference

Languages

English	Full Professional Proficiency	TOEIC: 980/990
French	Mother tongue	
Arabic	Mother tongue	
Spanish	Intermediate	
Russian	Beginner	

Interests

Sports	Boxing, Wrestling, Hiking.
Other	Reading.